

Constructing Test Cases Using Natural Language Processing

Prof. Ahlam Ansari

Department of Computer Engineering
M. H. Saboo Siddik College of Engineering
Byculla, India
ahlam@mhssce.ac.in

Ansari Sadaf Fatima

Department of Computer Engineering
M. H. Saboo Siddik College of Engineering
Byculla, India
ansarisadaffatima@gmail.com

Mirza Baig Shagufta

Department of Computer Engineering
M. H. Saboo Siddik College of Engineering
Byculla, India
shaguftamirzabaig@gmail.com

Shaikh Tehreem

Department of Computer Engineering
M. H. Saboo Siddik College of Engineering
Byculla, India
tehreem.shaikhrizwan@gmail.com

Abstract— Testing of product is perform to discover or detect the errors and defects in the developed system. But testing is usually time consuming especially when complex projects are canvass. Testing of a product lead off with generation of test cases. The Test case generation are based on three parts coding, design and specification. The Specification based testing deals with generation of test cases from the functional requirements. The proposed system deals with automatic generation of test cases from functional requirement using Natural Language Processing (NLP). The proposed system constructs test cases based on keywords in context from functional requirement of Software Requirement Specification document. The proposed system is beneficial as it can automatically analyze the functional requirement from Software Requirement Specification in order to extract test cases for testing. The goal of the proposed system is to reduce effort and time consumed by software tester to test the product.

Keywords—testing; test cases; functional requirements; Natural Language Processing (NLP); Software Requirement Specification;

I. Introduction

In the present era, firm faces a stiff competition so it becomes necessary to increase quality of a product. So they focus on research and development. Testing is required to show the standard of product in research and development. Therefore testing is considered as a very important stage of the product development life cycle.

The major purpose of testing is to discover the error in the product so as to ensure error free and reliable delivery of the product to the customer or client and to meet customer's satisfaction. Therefore a dedicated department has been set up in the organization for testing. Testing teams needs to create test cases in order to begin the testing process which consumes

maximum efforts and time. Test cases not only define what to test, what operations to invoke, in what order but also how to test the requirement and what test data values are needed. Considering this problem, the proposed system is based on the notion of automatic generation of test cases. It is focusing on specification based testing that deals with the generation of test cases from the functional requirements using Natural Language Processing. The proposed system will automatically analyze the requirement document based on keywords in context to extract the test cases for testing.

II. Literature Review

A. Software Requirement Specification

The SRS specifies requirements of the system. It provide how system will work perform in real time environment, its interaction with user and what is the outcome from the system. The SRS provides the complete description of the system. The SRS provides the complete description of the system. It includes the following:

1) Introduction

It includes document purpose, product scope, intended audience and document overview, definitions, acronyms and abbreviations, references and acknowledgments.

2) Overall Description

It includes product perspective, product functionality, users and characteristics, operating environment, design and implementation constraints, user documentation, assumptions and dependencies.

3) Specific requirements

It includes External interface requirements, functional requirements, Public Relation Department (PRD) tables and behavior requirements.

4) Other Requirements

It includes appendix which may include Terminology or Glossary or Definitions list which should be in alphabetical order.

B. Functional requirement

Requirements form the basis of any system's foundation. For any product development life cycle determining the right requirement for system is very critical it should be well thought out, clearly understood in order to not to be doomed to failure and hence emphasis on this phase should be given. As it is one of the most important stages in development lifecycle of any system it should be carefully implemented to have high quality and reliable system as an outcome [1].functional requirements shows the functionality and behavior of the system. It describes the system's input, output and relationship between them [2].

C. Natural Language Processing (NLP)

The effective way of communication is "language" for human. In the case of computer, the area of research and application which shows how computer is used to process and understand the natural language text and voice speeches. The main aim of NLP is to analyzing and understanding and base on gathered information it generates a language in both written and spoken form. NLP is nothing but the making the computer to understand how humans learn and uses language. The base of NLP has different area as follows:

- Computer science
- Linguistics
- Mathematics
- AI
- Robotics

D. Test case

In software engineering, test case is the process of testing features and all the requirements of applications. , the test case table structure is shown in TABLE I.

TABLE I. TEST CASE TABLE

Steps	Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)	Notes

III. Existing System

Tester spends a lot of time in generating test cases manually which is a very tedious task. There may also be possibility of missing some of the test cases. In its most basic form, a test case is just a set of steps in a script form that can be executed in order to reproduce the intended functionality of a feature within the software.

Once a tester is familiar with the features and functionality, they can create test cases to be used, when testing becomes necessary. The best time to write test cases for manual testing is when the system is still in the early stages of development. Once the functionality of the program

has been set in stone, the documentation can be used as a source for the test cases. Instead of looking at an early, unfinished version of the product, use clearly defined documentation of the end-goal functionality to avoid the creation of flawed test cases.

After successfully analyzing the requirement in Software Requirement Specification creating the test cases, the next step is to generate test cases. In this the tester uses his past experience and creates a list of test cases and then one after the other these test cases are implemented as in the list.

IV. Proposed System

Testing teams needs to develop test cases in order to start the testing process which consumes maximum efforts and time. Also there may be a possibility that tester might make mistake in properly analyzing the requirement and can miss some test cases.

The project mainly focuses at increasing the efficiency of testing by automatically generating test cases from functional requirement document based on keywords in context to extract the test cases for testing and to reduce the time, effort and cost require for testing.

a. Working of the system

The block diagram of constructing test cases using NLP is shown in below mention Fig. 1.

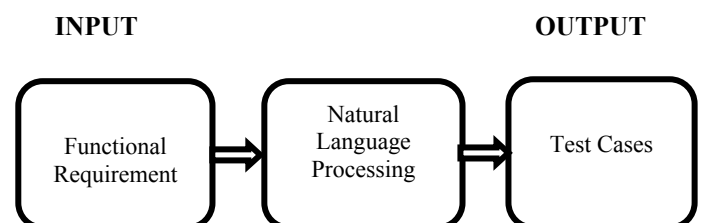


Fig. 1. Block Diagram for proposed system

Several techniques as mentioned in literature review have been proposed for automatic construction of test cases.

We are using three steps generally .First, we are providing functional requirement as input. Second, NLP is used as intermediate for processing and analyzing the functional requirement document and also extracting the required data for construction of test cases. Third, output is Test case table based on extracted data.

b. Algorithm

Input Set: Functional Requirement Document

Processing:

1. Start.
2. Feed the document to NLP.
3. Check each scenario.
4. Check for each cases within scenario.
5. Extract conjunctive statement containing 'if' and 'then' keyword.
6. Extract noun from conjunctive statement and put it into 'Test Data' field of test case table.
7. Put the statement associated with if in 'Test Steps' Field.
8. Put statement associated with 'then' in 'Expected Result' Field.
9. Compare actual result and expected result and update the 'Status' Pass/Fail.

if (actual result==expected result)

then Status="Pass";

else

{

Status="Fail";

Comment="reason for failure";

}

10. if (more conjunctive statement)

then go to step 5

else if (more cases)

then go to step 4

else if (more scenario)

then go to step 3

else

go to step 11

11. Stop.

c. Expected Results

Let's take the scenario of Functional requirement for the login page.

Case1: "Check if user name and Password is correct then login successfully else login Failure as Invalid user name and Password".

Case2: "Check if user name and Password is incorrect then login failure else login successfully as Validation Error".

Case3: "Check if user name correct and Password is incorrect then login failure else login successfully as Validation Error".

Case4: "Check if user name incorrect and Password is correct then login failure else login successfully as Validation Error".

Case5: "If user forgot the password then Go to reset password page else page not found error as server is down".

Case6: "If user perform three incorrect passwords attempts then Block the accounts for 15 minutes, Generate OTP and send it else Allow further attempts as Authentication failure".

Then the output will be generated as Test case table for login page as shown in TABLE II.

TABLE II. TEST CASE TABLE FOR LOGIN PAGE

STEPS	TEST STEPS	TEST DATA	EXPECTED RESULTS	ACTUAL RESULTS	STATUS (PASS /FAIL)	NOTES
1	user name and Password is correct	User name and password	login successfully	login successfully	PASS	-
2	user name and Password is correct	User name and password	login successfully	login Failure	FAIL	Invalid username and password
3	user name and Password is incorrect	User name and password	Login Failure	login Failure	PASS	-

4	user name and Password is incorrect	User name and password	Login Failure	login successfully	FAIL	Validation Error
5	username correct and password is incorrect	User name and password	Login Failure	Login Failure	PASS	-
6	username correct and password is incorrect	User name and password	Login Failure	login successfully	FAIL	Validation Error
7	username incorrect and password is correct	User name and password	Login Failure	Login Failure	PASS	-
8	username incorrect and password is correct	User name and password	Login Failure	Login successfully	FAIL	Validation Error
9	User forgot the password	Password	Got to reset password page	Got to reset password page	PASS	-
10	User forgot the password	Password	Got to reset password page	Page not found error	FAIL	Server is down
11	User perform three incorrect password attempts	Password	Block the accounts for 15 minutes, Generate OTP and send it	Block the accounts for 15 minutes, Generate OTP and send it	PASS	-
12	User perform three incorrect password attempts	Password	Block the accounts for 15 minutes, Generate OTP and send it	Allow further attempts	FAIL	Authentication failure

V. Future Scope

Currently proposed system only deals with constructing test cases from the functional requirement given in conjunctive statement format. Hence the system can be extended to automatically construct test cases from the functional requirement given in any form. So as to partially or to some extent completely eliminate the need of human intervention in the testing process. The accuracy of the system can be increased, if we could anyhow predict that what test cases would prove to be more important in the future, then their priorities could be uplifted. But practically, as of now, it is impossible to predict, in advance, that which test case might be more important in the future.

VI. Conclusion

This proposed has introduced an automation approach to generate test cases using functional requirements. Due to automation, maximum efforts and time consumed in developing test cases is being saved. Thereby reducing cost included in testing of projects. Maximum coverage of all requirements specified in SRS is ensured by this automation approach. Hence reliable system can be deployed to the customer that will truly satisfy customer's need.

ACKNOWLEDGMENT

Authors would like to express heartfelt and sincere gratitude to the principal **Dr. Mohiuddin Ahmed**, M. H. Saboo Siddik College of Engineering, Mumbai, for providing

the facilities to carry out the work. Authors also wish to express grateful thanks to beloved Head of the Department of Computer Engineering **Dr. Z. A. Usmani**, who gave us full support and constant encouragement, valuable suggestions and helping tendency which has made us to carry out and finish the work successfully. Finally, this work might not have been possible had it not been for the efforts of our internal project guide **Prof. Ahlam Ansari**, for valuable suggestions and constant encouragement for successful completion of the work.

REFERENCES

- [1] Ahlam Ansari, Moonish Maknojia and Altamash Shaikh, "Intelligent Information Extraction Based On Artificial Neural Network", Department of Computer Engineering, M.H. Saboo Siddik College of Engineering, Mumbai, India, January 2016. IJFCST.
- [2] Ahlam S.A.Ansari, Prof. Kailas K. Devadkar, Dr. Prachi Gharpure, "Optimization of Test Suite- Test Case in Regression Test", Department of Computer Engineering, Sardar Patel Institute of Technology, University Of Mumbai, Mumbai, India 2 Department of Information

Technology, Sardar Patel Institute of Technology, University Of Mumbai, Mumbai, India ,2013 IEEE.

- [3] Hécio A. Soares and Raimundo S. Moura, “A methodology to guide writing Software Requirements Specification document”, Departamento de Informática Instituto Federal do Piauí and Departamento de Computação Universidade Federal do Piauí, 2015 IEEE.
- [4] Test-case-template (26 September, 2016) Retrieved from <http://www.softwaretestinghelp.com/test-case-template-examples/>
- [5] Saad Alsaleh and Haryani Haron, “The Most Important Functional and Non-Functional Requirements of Knowledge Sharing System at Public Academic Institutions: A Case Study”, Lecture Notes on Software Engineering, Vol. 4, No. 2, May 2016.
- [6] J.Zalewski, “How to write the SRS documentation, following IEEE Std. 830.”, ISM 4331, J.Zalewski, September 2003.
- [7] Fatima Zahra Hammani ,“Survey of Non-Functional Requirements Modeling and Verification of Software Product Lines”, IMS Team, SIME Laboratory ENSIAS, Mohammed V Souissi University Rabat, Morocco, 2014 IEEE.
- [8] Harry M. Sneed and Chris Verhoef , “Natural Language Requirement Specification for Web Service Testing” , Anecon GmbH, Vienna and VU Amsterdam , 2013 IEEE.
- [9] Sun Rui and Zhong Deming , “Translating Software Requirement from Natural Language to Automaton” , Beihang University, Beijing, China , 2013 International Conference on Mechatronic Sciences, Electric Engineering and Computer (MEC)Dec 20-22, 2013, Shenyang, China, 2013 IEEE.
- [10] Milos Gligoric, Lamyaa Eloussi, and Darko Marinov , “Practical Regression Test Selection with Dynamic File Dependencies” , University of Illinois at Urbana-Champaign , ISSTA’15, July 13–17, 2015, Baltimore,MD, USA <http://dl.acm.org/citation.cfm?doid=2771783.2771784>.
- [11] Wei Jin, Alessandro Orso and Tao Xie , “Automated Behavioral Regression Testing” , Georgia Institute of Technology , Georgia Institute of Technology and North Carolina State University , 2010 Third International Conference on Software Testing, Verification and Validation, 978-0-7695-3990-4/10 \$26.00 © 2010 IEEE.
- [12] Árpád Beszédés, Tamás Gergely, Lajos Schrettnér, Judit Jász, László Langó, Tibor Gyimóthy , “Code Coverage-Based Regression Test Selection and Prioritization in WebKit” , University of Szeged, Department of Software Engineering & HAS Research Group on AI, Szeged, Hungary , 2012 28th IEEE International Conference on Software Maintenance (ICSM) , 978-1-4673-2312-3/12/\$31.00 c 2012 IEEE.
- [13] Sebastian Elbaum† , Gregg Rothermel† , John Penix‡ ,” Techniques for Improving Regression Testing in Continuous Integration Development Environments” , †University of Nebraska - Lincoln Lincoln, NE, USA , ‡Google, Inc. Mountain View, CA, USA , FSE’14, November 16–21, 2014, Hong Kong, China.
- [14] Kshirasagar Naik And Priyadarshi Tripathy , “Software Testing And Quality Assurance Theory and Practice” , Department of Electrical and Computer Engineering University of Waterloo, Waterloo and NEC Laboratories America, Inc., A John Wiley & Sons, Inc., Publication , 2008.
- [15] Roger Pressman , “Software Engineering A Practitioner's Approach 7th Edition” , McGraw-Hill, a business unit of The McGraw-Hill Companies, Inc., 1221 Avenue of the Americas, New York, NY 10020. Copyright © 2010 by The McGraw-Hill Companies, Inc.
- [16] Arjinder Singh and Er. Sumit Sharma , “Functional Test Cases Generation Based on AutomatedGenerated Use Case Diagram” , Chandigarh University , International Journal of Innovative Research in Advanced Engineering (IJIRAE) ISSN: 2349-2163 Issue 8, Volume 2 (August 2015) , © 2014-15, IJIRAE.
- [17] Ron Patton , “Software engineering”. 800 E. 96th St., Indianapolis, Indiana, 46240 USA.